

Power BI DAX Lab (Accounting) - Student Handout

Dataset: **PowerBI_DAX_Accounting_Student.xlsx**

Important for copy/paste: The DAX below assumes your Power BI model uses these exact table names: tblFactTransactions, tblDimAccount, tblDimDepartment, tblDimEntity, and tblCalendarRef. Load the Excel *tables* (not just sheets), or rename the imported model tables so the names match exactly before pasting the formulas.

Part A — Import + Model Setup

Import these tables from the workbook: **tblFactTransactions**, **tblDimAccount**, **tblDimDepartment**, **tblDimEntity**, and **tblCalendarRef**.

Create these star-schema relationships:

From (Fact)	To (Dimension)
tblFactTransactions[AccountNo]	tblDimAccount[AccountNo]
tblFactTransactions[DeptCode]	tblDimDepartment[DeptCode]
tblFactTransactions[EntityID]	tblDimEntity[EntityID]
tblFactTransactions[TransDate]	tblCalendarRef[Date]

Mark tblCalendarRef as the date table using the Date column.

Part B — Required Measures (exact DAX to copy/paste)

Net Amount

```
Net Amount = SUM(tblFactTransactions[Net])
```

Total Debit

```
Total Debit = SUM(tblFactTransactions[Debit])
```

Total Credit

```
Total Credit = SUM(tblFactTransactions[Credit])
```

Transaction Count

```
Transaction Count = COUNTROWS(tblFactTransactions)
```

Distinct Entities

```
Distinct Entities = DISTINCTCOUNT(tblFactTransactions[EntityID])
```

Revenue

```
Revenue = CALCULATE([Net Amount], tblDimAccount[AccountType] = "Revenue")
```

Expense

```
Expense = CALCULATE([Net Amount], tblDimAccount[AccountType] = "Expense")
```

Net Income

```
Net Income = [Revenue] - [Expense]
```

Profit Margin %

```
Profit Margin % = DIVIDE([Net Income], [Revenue])
```

Posted Net

```
Posted Net = CALCULATE([Net Amount], tblFactTransactions[Status] = "Posted")
```

YTD Net

```
YTD Net = TOTALYTD([Net Amount], tblCalendarRef[Date])
```

Net PY

```
Net PY = CALCULATE([Net Amount], SAMEPERIODLASTYEAR(tblCalendarRef[Date]))
```

YoY Change

```
YoY Change = [Net Amount] - [Net PY]
```

YoY %

```
YoY % = DIVIDE([YoY Change], [Net PY])
```

Material Net (>=5000)

```
Material Net (>=5000) = CALCULATE([Net Amount], FILTER(tblFactTransactions,  
ABS(tblFactTransactions[Net]) >= 5000))
```

Missing Dept Count

```
Missing Dept Count = COUNTROWS(FILTER(tblFactTransactions, tblFactTransactions[DeptCode] =  
""))
```

Extended Amount (SUMX)

```
Extended Amount (SUMX) = SUMX(tblFactTransactions, tblFactTransactions[Quantity] *  
tblFactTransactions[UnitPrice])
```

% of Total by Account

```
% of Total by Account = DIVIDE([Net Amount], CALCULATE([Net Amount], ALL(tblDimAccount)))
```

Vendor Rank

```
Vendor Rank = RANKX(ALL(tblDimEntity[EntityName]), [Net Amount], , DESC)
```

Top 5 Vendor Spend

```
Top 5 Vendor Spend = CALCULATE([Net Amount], TOPN(5, ALL(tblDimEntity[EntityName]), [Net  
Amount], DESC))
```

Part C — Required Visuals

1. Monthly Trend (line or clustered column): Axis = tblCalendarRef[YearMonth]; Values = [Net Amount]; add a slicer for tblFactTransactions[Status].
2. Income Statement Matrix: Rows = tblDimAccount[AccountType]; Columns = tblCalendarRef[YearMonth]; Values = [Net Amount].
3. Exception Cards: [Material Net (≥ 5000)], [Missing Dept Count], and [Transaction Count].
4. Top Vendors Bar Chart: Axis = tblDimEntity[EntityName]; Values = [Net Amount]; filter or show Top 5.

Part D — Student Questions

- Q1. What is [Net Income] for the full period?
- Q2. Why is [Posted Net] different from [Net Amount]?
- Q3. How many transactions are missing DeptCode? Why is that a reporting risk?
- Q4. Why do we need SUMX to compute [Extended Amount (SUMX)]?
- Q5. Which vendor has the highest net impact? Use [Vendor Rank] or the Top 5 visual.